

Quantum Mechanics III

Lecture 7

Quantum State Tomography

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$p_k = \text{Tr}(\rho E_k)$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$\rho(T) = TT^\dagger / \text{Tr}(TT^\dagger)$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$\ell(\rho) = \sum_k n_k \ln \text{Tr}(\rho E_k)$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Park, ch. 7; State Reconstruction Service guide; QDC tomography block